

COINPICKS · RESEARCH & EDUCATION

Bitcoin Technical Analysis Quant Pillar Report

The Tape Itself — Measured, Scored, and Mapped to the Pillar Standard

*Machine-computed from 16 years of daily price data (5,878 days, 2010–2026) through a 7-indicator scoring stack, halving-cycle analogs, and a power-law valuation model. Every number is reproducible from the scripts in the BTC-TA folder. **These numbers are computed, not judged, so they carry no draft/committed status of their own — they enter the master blend directly. Today's overall qualitative score set is being left as a DRAFT awaiting Alex's manual review this cycle (a novel Iran War Risk escalation — reports that the UAE was hit by Iranian missiles for the first time — unrelated to this quant pillar); this pillar's own output is unaffected by that review.***

COINPICKS QUANT PILLAR REPORT · THE QUANTITATIVE SLEEVE OF MY THESIS

Data through August 20, 2026 (BTC \$71,882.92) · Pipeline re-run & verified August 20, 2026 · Not Financial Advice

The Scores — August 20 Model Output

Computed August 20, 2026 (previous computed run August 16, 2026; no cycle ran Aug 17–19) — same standard as every qualitative pillar. Today's mapping moves every window: 1mo 41→45, 3mo 36→43, 6mo 45→52, 1yr 68→67, 3yr 71→70 — a sharp near-term improvement as the tape repaired, with a small pullback at the long end (see What Changed for why). Today's overall qualitative score set is being left as a DRAFT awaiting Alex's manual review (a novel Iran War Risk escalation, unrelated to this quant pillar); this quant pillar is computed fresh every run regardless and carries no draft/committed status of its own.

WINDOW	MODEL P	QUALITY	IMPACT	SIGNAL	CALL
1 month	45	0.75	8	-0.60	Mild Bear — the tape is repairing (weekly composite flipped to +9.9, Neutral) but the current bar hasn't closed yet; the 1mo cycle-analog is still 0-of-3 positive
3 months	43	0.70	8	-0.78	Mild Bear — still the softest window; 1-of-3 analog positive, median -21.7%
6 months	52	0.55	8	+0.18	Neutral — a genuine coin flip now, tilting up as 2-of-3 analogs turn positive
1 year	67	0.60	8	+1.63	Bull — the cycle turn; every prior cycle at this exact month was higher a year later (3-of-3)
3 years	70	0.50	8	+1.60	Bull — value + cycle

Impact: 8 (the tape aggregates every driver, but adds no causal force of its own) · **Formula:** Signal = ((P - 50) ÷ 50) × Quality × Impact · **Honesty cap:** P = 50 + 25 × (window score ÷ 100) — pure TA never claims odds more than 25 points from a coin flip, because its measured out-of-sample edge is modest. Math machine-verified.

THE THREE INPUTS, IN PLAIN ENGLISH

- P** — the odds the tape is *good for Bitcoin* by then. Near-term now much closer to a coin flip (weekly composite +9.9/100, agreement narrowed to 4-of-7 as the tape repairs — price has now moved above the 200-week average, +11.9%, though the actual weekly bar hasn't closed yet); long-term still high (at this exact point in prior cycles, the next 12 months were positive 3-for-3, and price sits 48% below its power-law fair value).
- Quality** — how trustworthy the evidence is. Highest near-term (841 weekly bars of history feeding the indicator stack), lowest at 3 years (only 3 completed cycles to learn from, and this cycle is already breaking the old pattern).
- Impact** — 8 out of 10, now the uniform value shared by every pillar; the TA pillar carries an equal 20% weight in the master, same as each qualitative pillar. Price is the thing itself — but a mirror, not an engine.

The Claim

The thesis, and where the numbers differ from the mood

The crowd: -50% from the all-time high just weeks ago, a lower low on June 30, a weekly close under the 200-week average as recently as June 28 — "the cycle is dead, get out." (Price has since closed at \$71,882.92, up **+14.2% versus Aug 16** and +22.8% off the June 30 low, even as the qualitative score set for today sits in draft awaiting Alex's review over an unrelated geopolitical shock — the tape moving hard on its own schedule while the qualitative story is a war-risk headline is itself information: this move is coming from price action, not from any of the four qualitative pillars.)

The model: the short-term tape has repaired sharply — the weekly and monthly composites both flipped from outright bearish to Neutral (weekly $-29.0 \rightarrow +9.9$, monthly $-41.9 \rightarrow -12.8$), and price is above the 200-week average for the first time since the correction began. But the current weekly bar hasn't closed yet, and the 1–3 month windows are still Mild Bear (P 45/43) — this is not yet a "the correction is over" report. The 6-month window is now a genuine coin flip (P 52). The long read holds its direction but ticked down a point at each end (1yr $68 \rightarrow 67$, 3yr $71 \rightarrow 70$): every prior cycle at this exact month was higher 12 months later, price is still 48% below fair value, and the froth that precedes real crashes remains absent.

The edge: the crowd reads one clock (the tape) — and today, for once, the tape genuinely agrees with the crowd's improving mood. The model still reads three — tape (repairing, not yet confirmed), cycle clock (bottoming window: prior cycles bottomed at months 26–30; we're at month 28.02, with the average-bottom date landing almost exactly now), and valuation (48% below the power-law fair line, though price has now moved back above the -1σ "cheap" band for the first time this correction). All three clocks are pointed the same direction on the medium-to-long horizon now — a genuine change from mid-August, when the tape was the lone holdout.

The Facts — The Tape

Computed August 20, 2026 from Bitstamp daily data (BTC \$71,882.92, daily close as of August 20). Current weekly/monthly/quarterly bars are still forming — scores use them as-is. Quantified: on completed bars only, the weekly composite is -29.5 , monthly -32.6 , quarterly $+7.7$ — monthly and quarterly are essentially unchanged from Aug 16 (only their still-forming current bars moved); weekly's completed-bar figure actually moved slightly MORE bearish ($-25.0 \rightarrow -29.5$). Nearly all of today's headline improvement below lives in the still-forming current bars, not yet in confirmed closed bars.

- **Weekly composite $+9.9/100$ (Neutral), agreement narrowed to 4-of-7 indicators.** Price is now above the 20-week (\$69,205, $+3.9\%$) and, for the first time this correction, above the **200-week average (\$64,257, $+11.9\%$)** — the headline development this cycle (see below) — but remains well below the 50-week (\$81,680, -12.0%); RSI 51.0; MACD line still negative ($-4,053$) with histogram positive ($+1,017$) — momentum repair continuing; %B 0.60.
- ★ **Repair signal #1 (a weekly close above the 200-week MA) is now trending strongly toward confirmation — but is not yet confirmed.** Price closed Aug 20 at \$71,882.92, $+11.9\%$ above the 200-week average (\$64,257) on a daily basis — a dramatic reversal from Aug 16, when price sat fractionally below it. The actual weekly bar that confirms this signal is still forming (dated 2026-08-23) and doesn't close until Sunday; until then this is a strong lean, not a confirmed close. Worth flagging honestly: the completed-bar weekly composite (prior closed bars only, no partial data) is -29.5 — more bearish than Aug 16's -25.0 — so today's improvement is concentrated almost entirely in the current, still-forming bar.
- **Monthly composite $-12.8/100$ (Neutral), agreement narrowed to 3-of-7** — a sharp improvement from Aug 16's unanimous 7-of-7 bearish reading. Below the 20-month average (\$87,887, -18.2%), above the 50-month (\$60,918, $+18.0\%$); RSI 48.0; monthly MACD line -330 , histogram $-5,201$ — still negative but far less deeply so; %B 0.27. The completed-bar monthly composite is -32.6 , unchanged from Aug 16 — like the weekly reading, this month's improvement is concentrated in the still-forming current bar (dated 2026-08-31).
- **Quarterly composite $+18.5/100$ (Neutral), agreement 3-of-7** — firmed further from Aug 16's $+8.3$. Comfortably above the 20-quarter average (\$57,816, $+24.3\%$) and the 50-quarter (\$28,603, $+151.3\%$); RSI 55.6, MACD line $+15,697$ but histogram -857 (momentum still cooling). Completed-bar quarterly composite is $+7.7$, unchanged from Aug 16 (current bar dated 2026-09-30, still forming).
- **Levels that matter:** resistance \$89,164 ($+24.0\%$, 1 touch), then \$81,685 ($+13.6\%$, 2x touch), then \$74,434 ($+3.5\%$, 1 touch) — price (\$71,883) is now trading just under that nearest resistance. Support \$59,930 (-16.6% , 1 touch), then \$57,735 (-19.7% , 1 touch); the falling weekly support trendline sits at \$45,626, the falling resistance trendline at \$77,700.
- ★ **June 30 remains the cycle low: \$58,526 close** — the flush tagged the Fib golden pocket (\$54,257–\$57,802 = 61.8–65% retrace of the \$15,479 $\rightarrow$ \$126,272 run) and has since built a **$+22.8\%$ bounce to \$71,882.92. \$58,526 is still the line in the sand** — a decisive break back below it would undo this entire recovery leg.



Weekly tape, scored +9.9/100 with agreement narrowed to 4 of 7 indicators as the tape repairs: price is now above the 20- and 200-week averages, still below the 50-week, MACD histogram flipped positive. Machine-annotated from the same data that feeds the scores.

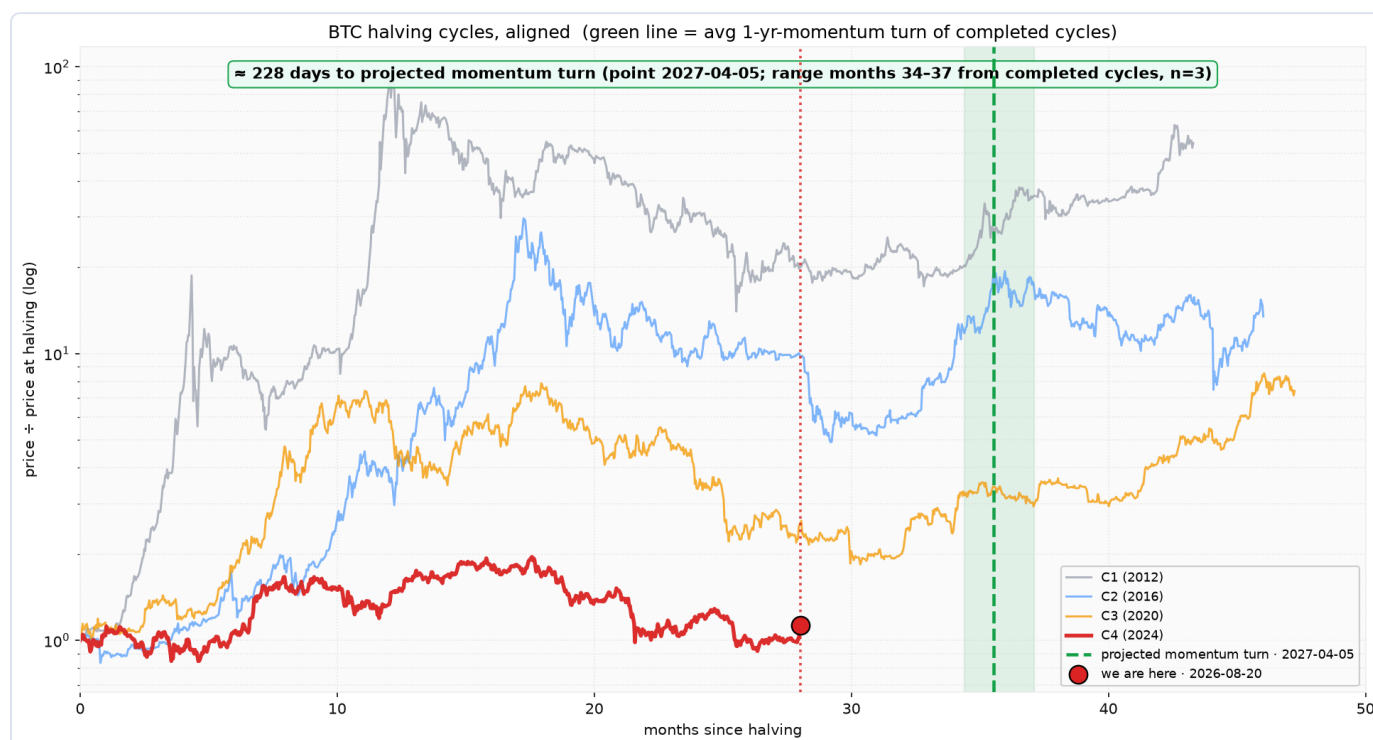
The Facts — The Cycle Clock & The Value Floor

The two slower clocks. Same data, longer lenses.

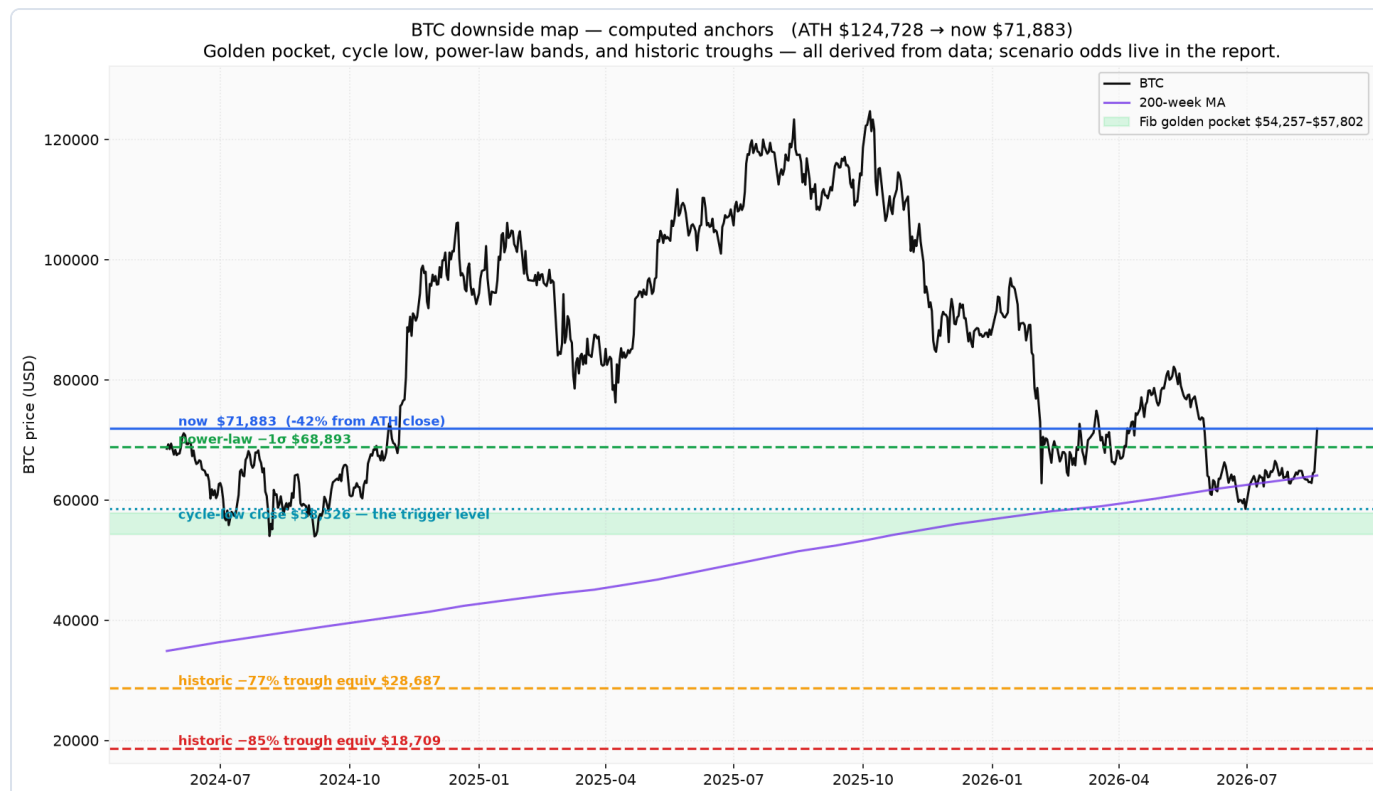
- The halving clock says "bottoming window."** The three completed cycles topped 12–18 months after their halving (avg 15.8) and bottomed at months 26–30 (avg 28.4). This cycle topped at month 18 (Oct 6, 2025 — \$124,728) and printed its low-so-far at month 26 (June 30, \$58,526, –53% from the top). **We are at month 28.02 — essentially at the average-bottom date (~Aug 31, 2026) itself.**
- ★ The cycle analogs are the spine of the long-window scores — and drifted again this cycle, a real fragility worth flagging.** Standing at this exact cycle-month in C1/C2/C3: next 1 month positive **0-of-3** (median –12.7%) · next 3 months positive **1-of-3** (median –21.7%, a worse median than Aug 16's, despite an extra cycle now agreeing) · next 6 months positive **2-of-3** (median +0.3%, up from 1-of-3) · next 12 months positive **3-of-3** (median +39.3%, down from +45.6%) · next 36 months positive **3-of-3** (median +968.1%, up from +859.8%). *Fragility disclosure, live example: several of these medians — and even two agreement counts (3-month, 6-month) — moved as the cycle-month clock advanced from 27.9 to 28.02, exactly the kind of swing this report has warned $n=3$ analogs are prone to. Treat the analog levels as soft, the direction as the signal. And know what the 36-month number is: from month ~28 that window reaches month ~64 — it contains the NEXT halving's rally.*
- 1-year momentum turned positive at months 34–37 in every completed cycle (avg 35.5) → still projects to ~April 2027** (80% window Jan–Jul 2027) for this cycle.
- The power-law floor:** a log-log fit over the full 16-year series puts fair value at **\$138,020** — price is **–48%** versus fair. Fit-window honesty: starting the fit in 2011, 2012, 2013, or 2015 instead moves fair value to \$127,938 / \$132,918 / \$116,482 / \$143,366 respectively (fair value ranges roughly \$116K–\$143K depending on fit window) — the level is soft, but the **"well below fair" verdict holds across every window tested.** One genuine change

worth flagging: price (\$71,882.92) has now moved back **above the -1σ band (\$68,893)** — out of the "cheap" zone for the first time this correction (the -2σ band sits at \$34,388, far below). Every halving-era cycle trough was -77% to -85% from ATH (\$18.7K–\$28.7K equivalent today) — the pre-halving 2011 bear went $\sim -93\%$ ($\sim \$8.7K$ equivalent), excluded from the range but in the data; this cycle's low-so-far was -53% from the ATH.

- **No froth in the system:** the altcoin-froth index reads $z = -0.87$ — the opposite of the euphoric signature that preceded prior deep crashes. Validated on 16 years of data: high-froth days led to -39% average 12-month drawdowns vs -23% for low-froth days (gap significant, 95% CI excludes zero). *Data lags ~ 6 weeks (through May 23, unchanged)* — used as a tilt, not a score input.
- ★ **What we rejected:** a 5-feature logistic crash model was tested leave-one-cycle-out and **failed** (AUC 0.44 vs 0.58 for froth alone; 0.70 = useful). It is not in this report's numbers. The scores rest only on what survived out-of-sample testing.



All four halving cycles aligned (log scale, price ÷ price-at-halving). Triangles mark where 1-year momentum turned positive in each prior cycle — months 34–37, projecting to $\sim$ April 2027 for this one. The red dot is today: month 28.02, the historical bottoming window.



The downside map, updated after the June 30 flush: the projected \$54K–\$61K bottom zone was hit (\$58,526 close / \$57,735 intraday, inside the golden pocket's doorstep) and bounced. Odds re-weighted accordingly: ~40% the low is in, ~35% one more flush to \$54–\$58K, ~17% deeper bear, ~8% tail.

The Prize

Illustrative only — not price targets, not advice

From \$71,882.92: the 20-week average (~\$69.2K) is already behind us — price sits **+3.9%** above it, the first leg of trend repair; the 50-week average (\$81,680) is the next test, **+13.6%** away. A retest of the \$124,728 ATH is **+73.5%**; the power-law fair line is **+92.0%**. The median 12-month analog from this cycle-month is **+39.3%** (~\$100.1K by mid-2027); the median 36-month analog is +968.1% — **discount that one hard**, it leans on regimes (2017 mania, post-COVID ZIRP) that don't repeat on demand.

The Prize doesn't feed the math — it tells you what the lean is worth acting on. Here it says: the payoff for holding through a still-soft quarter (1mo/3mo remain Mild Bear) is a full cycle turn — and today's tape read makes that holding period look less lonely than it did four days ago.

The Other Side

The strongest honest case the model is wrong

- **n = 3.** Every long-window claim rests on three completed cycles. Three coin flips landing heads is not a law of nature.
- **This cycle already broke the pattern.** A -53% drawdown at month 26 vs -77/-85% in prior cycles could mean dampened cycles (bullish read) — or it could mean *the bottom isn't in*: a historic-depth trough from this ATH is \$18.7K–\$28.7K, far below every level in this report.
- **The ETF era changed the machine.** Prior cycles had no spot-ETF flow channel, no CME basis trade at this scale, no corporate treasuries. Analogs may simply not transfer.

- **The power law is a curve fit, not a promise.** R^2 0.96 in-sample says nothing about a structural break — a fit like this "said cheap" through most of 2022's decline too. It bounds scenarios; it doesn't time anything.
- **The tape's edge is small by construction.** The honesty cap exists because measured predictive validity is modest (froth AUC 0.58). If you removed the cap, this report would be louder — and less true.

The Triggers

Price and dated events — some prove the read right, some prove it wrong

WHEN / LEVEL	EVENT	MEANS
\$58,526	Daily close below the June 30 low	Flush scenario live — golden pocket \$54.3–57.8K is the target; re-rate 1/3-month down
~\$64,257	Weekly close above the 200-week MA	Trending strongly toward confirmation — the daily close (Aug 20) already sits +11.9% above this level; the actual weekly bar (dated 2026-08-23) doesn't close until Sunday, which is now the real remaining test
~\$69,205	Weekly close above the 20-week MA	Daily close is already clear (+3.9%); a confirmed Sunday weekly close above it would lock in trend repair — re-rate 1/3-month up if it holds
~Aug 31, 2026	Historical average-bottom month (28.4) passes	Arriving in ~11 days with price already well off the June 30 low; a higher low + improving monthly momentum after this date = basing confirmed
Jan–Jul 2027	Projected 1-yr momentum turn window (point: ~Apr 2027)	Arriving without new lows = cycle thesis on track; re-rate 1-year up
\$50,000	Weekly close below	–60% from ATH; the dampened-cycle story is failing — see the flip line

The flip line: a decisive weekly close below **\$50,000** breaks the base case and the golden pocket both — treat the dampened-cycle thesis as broken and put the historic trough zone (\$18.7K–\$28.7K) back on the table. Above it, a bearish quarter is the plan, not a surprise.

My Decision

QUALITATIVE SET: DRAFT — AWAITING ALEX'S MANUAL REVIEW · THIS PILLAR: COMPUTED AUGUST 20, 2026 (NO DRAFT/COMMITTED STATUS OF ITS OWN)

Computed P: 45 / 43 / 52 / 67 / 70 · Quality: 0.75 / 0.70 / 0.55 / 0.60 / 0.50 · Impact: 8

This is the quantitative sleeve of the thesis — the same five windows, same formula, same table as every qualitative pillar, but computed instead of judged. The near-term read has improved sharply today: the tape composites flipped from bearish to Neutral and price cleared the 200-week average on a daily basis, though the weekly bar itself hasn't closed and the 1–3 month windows are still Mild Bear. The 6-month window is now a genuine coin flip (P 52). The long read ticked down slightly at both ends (1yr 68→67, 3yr 71→70) even as everything short-term improved — a reminder that these windows partly price in how much of the value/analog gap has already closed, so a rally can pull long-window P down a touch even while the near-term score rises.

These numbers enter the master blend exactly as computed — this pillar has no override/re-stamp step of its own. Unlike the four qualitative pillars, this pillar carries no draft/committed status of its own: today's overall qualitative score set is being left as a DRAFT awaiting Alex's manual review — not for any reason in this report, but because of a novel Iran War Risk escalation (reports that the UAE was hit by Iranian missiles for the first time). This pillar's own computed output is unaffected by that review and enters the master blend the same way it does every day, draft or committed.

What Changed Since Last Check

Pipeline re-run August 20, 2026 (previous computed run August 16, 2026; no cycle ran Aug 17–19).

- ★ **Price surged: \$62,937.03 (Aug 16) → \$71,882.92 (Aug 20), +14.2%.** June 30's \$58,526 remains the cycle low; price is now +22.8% off it.
- ★ **Repair signal #1 flipped: price moved from fractionally BELOW the 200-week average to +11.9% above it.** On Aug 16 the daily close sat –1.7% under the 200-week MA (\$63,996); today's close (\$71,882.92) sits +11.9% above the updated 200-week MA (\$64,257) on a daily basis. This is the headline development of the cycle so far — but the actual weekly bar (dated 2026-08-23) hasn't closed yet, so it is trending strongly toward confirmation, not yet confirmed; Sunday's close is the real remaining test.
- **Composites flipped from bearish to Neutral on the current (forming) bar; completed-bar figures barely moved.** Weekly –29.0 → +9.9 (agreement narrowed 5-of-7 → 4-of-7), monthly –41.9 → –12.8 (agreement narrowed 7-of-7 → 3-of-7), quarterly +8.3 → +18.5 (agreement ~3-of-7). But on completed bars only: weekly –25.0 → –29.5 (slightly MORE bearish), monthly –32.6 → –32.6 (unchanged), quarterly +7.7 → +7.7 (unchanged) — nearly all of today's headline tape improvement lives in the still-forming current bars, not in confirmed closed bars. Worth taking seriously, but not yet a confirmed regime change.
- **Model P: 41/36/45/68/71 → 45/43/52/67/70.** The three near-term windows all moved up meaningfully (1mo +4, 3mo +7, 6mo +7) on the tape repair. The two long windows each ticked DOWN a point (1yr 68→67, 3yr 71→70) even as everything else improved — an honest artifact of the mapping: long-window P partly reflects how much of the value/analog gap remains open, and today's rally closed some of that gap (price moved from roughly –54% to –48% vs. power-law fair value, and out of the –1σ "cheap" band), leaving slightly less "already-cheap" edge to price in at the long end.
- **Cycle analogs drifted:** 3-month agreement improved from 0-of-3 to 1-of-3 positive, 6-month from 1-of-3 to 2-of-3 positive, as the cycle-month clock advanced from 27.9 to 28.02 — flagged per this report's standing n=3 fragility disclosure, not treated as new structural information.

- **Data refreshed:** Bitstamp daily through August 20, 2026 (5,878 days on the master series); Coin Metrics altcoin set still through May 23 (community data lags ~6 weeks — froth is a tilt, not a score input).
- **Context:** today's overall qualitative score set is being left as a DRAFT awaiting Alex's manual review (a novel Iran War Risk escalation — reported Iranian missile strikes on the UAE for the first time — unrelated to this quant pillar); this pillar's own computed output is independent of that process and enters the master blend the same way regardless of the qualitative set's draft status.

Method & Sources

Everything is reproducible — the scripts and data live in the BTC-TA folder. Run order below.

- **Price data:** Bitstamp OHLC API (daily, 2011–2026, no key) via [fetch_btc_history.py](#) · Coin Metrics community reference price (2010–2011 close-only) via [build_master_history.py](#) — 5,835 days total, each row tagged by source.
- **Indicator stack:** [indicators.py](#) — SMA 20/50/200, EMA 12/26, RSI-14 (Wilder), MACD 12/26/9, Bollinger 20/2, OBV — on weekly (W-SUN), monthly, quarterly bars. [levels.py](#) — swing-cluster S/R + least-squares trendlines.
- **Scoring:** [score.py](#) — each indicator → conviction in [−100, +100] (bounded inputs mapped linearly, unbounded squashed with tanh); composite = fixed weights (MA .20, RSI .15, MACD .15, BB .15, S/R .15, OBV .10, TL .10); agreement = indicators sharing the composite's sign.
- **Cycle & value:** [cycle_analysis.py](#) — halving-aligned tops/bottoms/momentum turns, projections from C1–C3 · [downside.py](#) / [downside_map.py](#) — power-law bands (R^2 0.96) + historical troughs + 6-method bottom-zone synthesis · [test_altcoin_froth.py](#) — froth quartiles vs forward drawdowns, block-bootstrap CI.
- **The mapping (new):** [pillar_mapping.py](#) — window scores from tape composites + cycle analogs + power-law value; $P = 50 + 25 \times (S/100)$; Quality rubric per window; writes data/pillar_scores.json for the master blend.
- **Run order:** fetch_btc_history → build_master_history → fetch_altcoin_data → indicators → levels → score → charts → cycle_analysis → downside → downside_map → pillar_mapping.
- **Honesty notes:** current weekly/monthly/quarterly bars are partial as of Aug 20 (completed-bar composites: weekly −29.5 / monthly −32.6 / quarterly +7.7 — monthly and quarterly are unchanged from Aug 16, only weekly's completed-bar figure moved, and it moved more bearish; nearly all of today's current-bar improvement is still "forming bar" noise until bars close) · the multi-signal crash model failed out-of-sample (AUC 0.44) and is excluded · the AUC figures are the pipeline's own attestation · froth data lags ~6 weeks · $n=3$ on all cycle claims, and analog windows can cross into the next cycle — the 6-month analog remains fragile on pure alignment drift, printed in the Facts · the 2024 halving is dated Apr 19 (US ET; block 840,000 hit Apr 20 00:09 UTC — 0.03 cycle-month effect) · S/R levels are algorithmic swing-clusters with 1–2 touches each · the bottom-odds split (40/35/17/8) is a judgment synthesis, not a model output.